EDPRE – Part B Project Report

Problem 5b: NOR Gate using MOS Transistors

Group Name	Construct and simulate a NOR gate using MOS transistors, infer its behavior, and explain its truth table with logical operation
Group Number	49
Group Members	Ritesh Maurya, Shikhar Sharma, Shubhika Verma, Kunjal
GitHub Link	[Paste your group GitHub link]

1. Title of Project

Construction and Simulation of a CMOS NOR Gate Using MOS Transistors, with Inference of Its Behavior and Truth Table.

2. Objective of Project

To construct and simulate a two-input NOR gate using p-channel and n-channel MOSFETs in Tinkercad, observe its behavior for every input combination using an LED as the output indicator, and explain the resulting truth table with the logical operation of the circuit.

- Build the pull-up (PMOS) and pull-down (NMOS) networks of a CMOS NOR gate.
- Apply logic inputs using two slide switches and read the output on an LED.
- Verify the truth table $Y = (A + B)'$ and explain why each state occurs.

3. Components and Tools Used

Name	Quantity	Component
T1, T2	2	pMOS Transistor (MOSFET)
T3, T4	2	nMOS Transistor (MOSFET)
BAT1	1	9 V Battery
D1	1	Red LED
R1	1	1 kΩ Resistor
S1, S2	2	Slideswitch

Tools:

- Tinkercad Circuits – circuit construction and simulation
- GitHub – hosting the simulation link and project documentation

4. Circuit Diagram

The circuit is a static CMOS NOR gate. The two PMOS transistors (T1, T2) are connected in series between the 9 V supply and the output node, and the two NMOS transistors (T3, T4) are connected in parallel between the output node and ground. Each input drives the gate of one PMOS and one NMOS transistor. The output node feeds the LED through the 1 kΩ current-limiting resistor to ground.

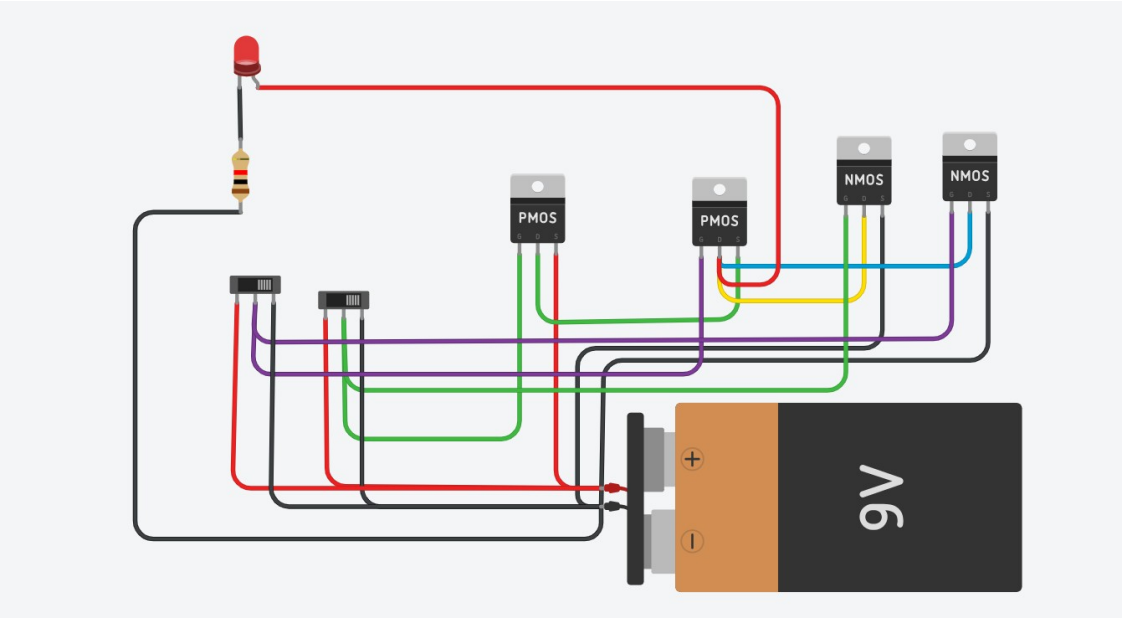


Figure 1: Tinkercad circuit of the CMOS NOR gate (both inputs LOW, LED ON)

Connections

Element	Connection
T1 (PMOS)	Source to +9 V; gate to input A; drain to source of T2
T2 (PMOS)	Source from drain of T1; gate to input B; drain to output node Y
T3 (NMOS)	Drain to Y; gate to input A; source to GND
T4 (NMOS)	Drain to Y; gate to input B; source to GND
S1, S2 (slide switches)	Outer terminals to +9 V and GND; middle terminal gives logic level to the gates of A and B
D1 and R1	Y → LED anode; LED cathode → 1 kΩ resistor → GND

5. Explanation

5.1 Circuit working

Each slide switch connects its gate line either to +9 V (logic 1) or to GND (logic 0). A PMOS transistor conducts when its gate is LOW and an NMOS transistor conducts when its gate is HIGH, so the two networks are complementary: exactly one of them connects the output to a supply at any time.

- **Pull-up network (series PMOS):** Y is connected to +9 V only if both T1 and T2 conduct, i.e. A = 0 AND B = 0.
- **Pull-down network (parallel NMOS):** Y is connected to GND if T3 OR T4 conducts, i.e. A = 1 OR B = 1.

Hence Y = 1 only when both inputs are 0, and Y = 0 otherwise, which is the NOR function: $Y = (A + B)' = A' \cdot B'$.

5.2 Truth table

A	B	PMOS T1 / T2	NMOS T3 / T4	Y (LED)
0	0	ON / ON	OFF / OFF	1 (ON)
0	1	ON / OFF	OFF / ON	0 (OFF)
1	0	OFF / ON	ON / OFF	0 (OFF)
1	1	OFF / OFF	ON / ON	0 (OFF)

Case-wise explanation

- **A = 0, B = 0:** Both PMOS are ON, so the series path from 9 V to Y is complete. Both NMOS are OFF, so no path to ground exists. Y is HIGH and the LED glows.
- **A = 0, B = 1:** T2 is OFF, breaking the series pull-up path. T4 is ON and pulls Y to ground. Y is LOW and the LED stays OFF.
- **A = 1, B = 0:** T1 is OFF, breaking the pull-up path. T3 is ON and pulls Y to ground. Y is LOW and the LED stays OFF.
- **A = 1, B = 1:** Both PMOS are OFF and both NMOS are ON. Y is pulled firmly to ground and the LED stays OFF.

5.3 Inference

The LED glows only when both switches are in the LOW position, which matches the NOR truth table. Because the pull-up and pull-down networks never conduct together, there is no direct path from supply to ground in steady state, so a CMOS gate draws almost no static power. Only the LED branch draws current when the output is HIGH; assuming an LED drop of about 2 V, this is roughly (9 − 2) V / 1 kΩ ≈ 7 mA. NOR is also a universal gate, so any other logic function can be built from it.

6. Simulation Circuit

The circuit was simulated in Tinkercad Circuits. The slide switches were toggled to apply the input combinations, and the LED state was observed. Screenshots of the simulated cases are shown below.

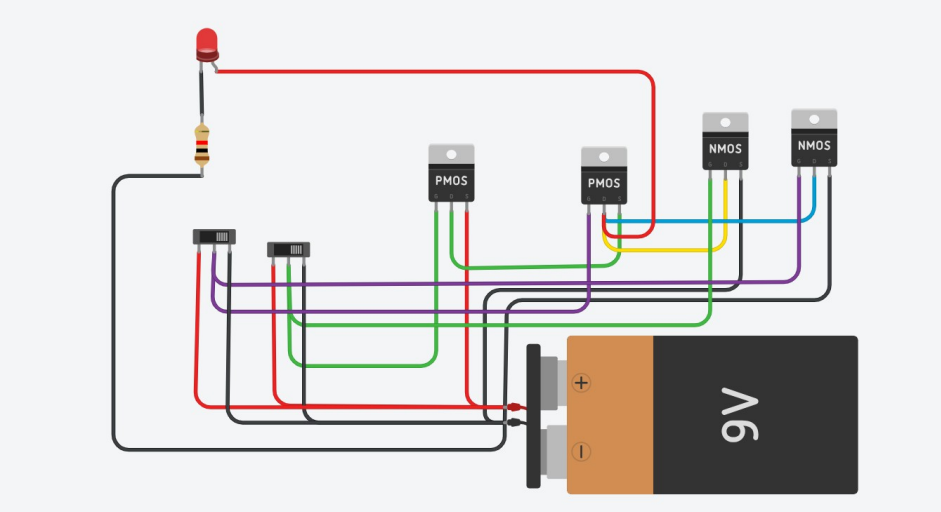


Figure 2: A = 0, B = 0 → Y = 1, LED glows brightly

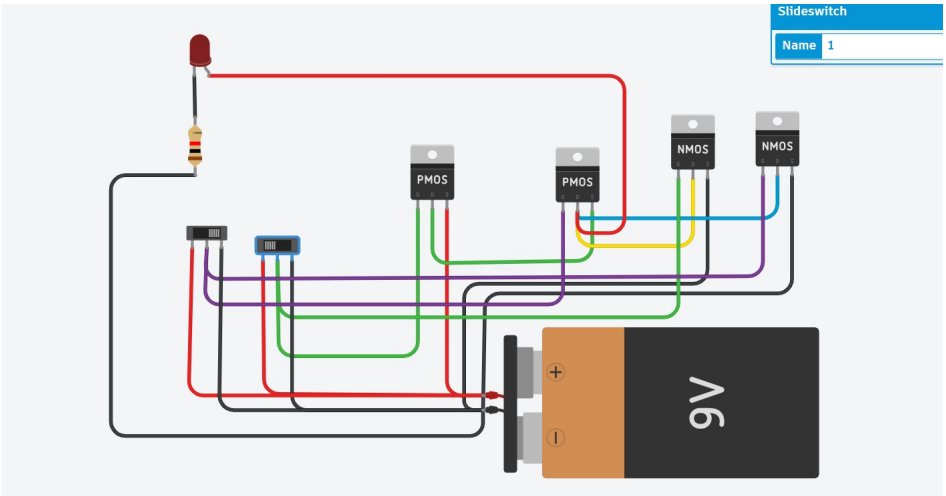


Figure 3: Switch S1 changed to HIGH → Y = 0, LED OFF

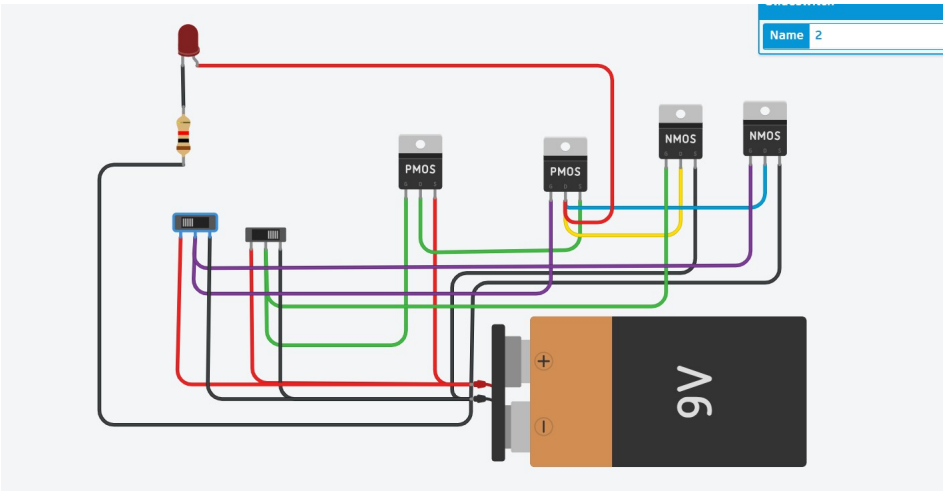


Figure 4: Switch S2 changed to HIGH → Y = 0, LED OFF

Tinkercad simulation link: Open Tinkercad circuit

https://www.tinkercad.com/things/6bDCYycMoJB/editel?returnTo=%2Fdashboard%3Fsource%3Dpwa&sharecode=ec9hMTDP9gIzEpWH47g2vEw4MmLnK7kjYbEmvxz6HBc

7. Code

The Tinkercad circuit is built purely from components and needs no microcontroller program. For completeness, the equivalent switch-level Verilog model of the same transistor network is given below.

```
module nor_cmos (input a, b, output y);
  supply1 vdd;
  supply0 gnd;
  wire mid;                  // node between the two series PMOS

  // Pull-up network: two PMOS in series (VDD -> mid -> y)
  pmos p1 (mid, vdd, a);
  pmos p2 (y,   mid, b);

  // Pull-down network: two NMOS in parallel (y -> GND)
  nmos n1 (y, gnd, a);
  nmos n2 (y, gnd, b);
endmodule

module tb;
  reg a, b;
  wire y;
  nor_cmos dut (a, b, y);
  initial begin
    $monitor("A=%b B=%b  Y=%b", a, b, y);
    a=0; b=0; #10;
    a=0; b=1; #10;
    a=1; b=0; #10;
    a=1; b=1; #10;
    $finish;
  end
endmodule
```

Expected output: Y = 1 for A=0, B=0 and Y = 0 for the other three combinations.

8. Conclusion

A two-input NOR gate was successfully constructed using two series PMOS and two parallel NMOS transistors and simulated in Tinkercad. The LED output confirmed the NOR truth table: the output is HIGH only when both inputs are LOW.